

Phased Application of Security in an Operational Environment in IoT

1. Design Phase

- Security by design approach
- Threat modeling and risk assessment
- Use secure hardware and protocols (e.g., TLS, MQTT-S)
- Embed secure boot and firmware

2. Development Phase

- Follow secure coding practices (OWASP IoT Top 10)
- Avoid hardcoded credentials
- Use secure libraries and perform code analysis

3. Deployment Phase

- Secure device provisioning and registration
- Enforce device authentication
- Secure communication setup
- Change default credentials

4. Operational Phase

- Real-time monitoring and anomaly detection
- Enforce access controls and policies
- Apply patches and firmware updates (OTA)
- Periodic vulnerability scanning

5. Decommissioning Phase

- Revoke device certificates and credentials
- Wipe sensitive data from devices
- Notify systems of device removal